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1 import numpy as np
2
3
4 def mse_loss(y_true,y_pred):
5     diff=y_true-y_pred
6     sq_diff = np.square(diff)
7     loss=np.mean(sq_diff)
8
9     return loss
10
11 def cross_entropy_loss(y_true,y_pred):
12     epsilon=1e-15
13     # F.binary_cross_entropy_with_logits
14     # nn.BCEWithLogitsLoss
15     y_pred=np.clip(y_pred,epsilon,1-epsilon)
16     loss=-np.mean(y_true*np.log(y_pred)+(1-y_true)*np.log(1-y_pred))
17     return loss
18
19 def multi_cross_entropy_loss(y_true,y_pred):
20     epsilon=1e-15
21     # nn.CrossEntropyLoss
22     y_pred=np.clip(y_pred,epsilon,1-epsilon)
23     loss = -np.mean(np.sum(y_true * np.log(y_pred), axis=-1))
24     return loss
25
26 if __name__ == "__main__":
27     batch_size=32
28     length=10
29     y_true=np.random.randn(batch_size,length)
30     y_pred=y_true+np.random.randn(batch_size,length)*0.01
31     loss=mse_loss(y_true,y_pred)
32     loss2=cross_entropy_loss(y_true,y_pred)
33     loss3=multi_cross_entropy_loss(y_true,y_pred)
34     print("MSE loss:",loss)
35     print("Cross Entropy Loss:",loss2)
36     print("Muliti Class Cross Entropy Loss:",loss3)

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